

B Simple Interest & Compound Interest -2

15/8/25

Ques

16) The difference b/w Simple Interest & Compound Interest on a certain sum of money for three years at 10% per annum is Rs 15.50. The sum is.

$$(SI - CI)_{3 \text{ yrs}} = \frac{P \times R^2 (300 + R)}{(100)^3}$$

W.K.T

$$CI_3 = 3a + 3b + c$$

$$a = \frac{PR}{100}, \quad b = a \left(\frac{R}{100} \right), \quad c = b \left(\frac{R}{100} \right)$$

$$CI_3 = 3 \left(P \frac{R}{100} \right) + 3 \left(\frac{PR}{100} \right) \left(\frac{R}{100} \right) + \left(\frac{PR}{100} \right) \left(\frac{R}{100} \right) \left(\frac{R}{100} \right)$$

$$= \frac{PR}{100} \left(3R + 3P \left(\frac{R}{100} \right) + \left(\frac{R}{100} \right)^2 \right)$$

$$= \frac{PR}{100} \left(3 + \frac{3R}{100} + \left(\frac{R}{100} \right)^2 \right)$$

$$SI_3 = \frac{PRT}{100} = \frac{3PR}{100}$$

$$(SI - CI)_3 = \frac{3PR}{100} - \frac{PR}{100} \left(3 + \frac{3R}{100} + \left(\frac{R}{100} \right)^2 \right)$$

$$= \frac{PR}{100} \left(\cancel{3} - \cancel{3} - \frac{3R}{100} - \left(\frac{R}{100} \right)^2 \right)$$

$$= - \frac{PR}{100} \left(\frac{3R}{100} + \left(\frac{R}{100} \right)^2 \right)$$

$$= - \left(\frac{PR}{100} \right) \left(\frac{R}{100} \right) \left(3 + \frac{R}{100} \right)$$

$$= - \left(\frac{155}{100} \right) \left(\frac{10}{100} \right) \left(3 + \frac{10}{100} \right)$$

$$P = 15.5$$

$$R = 10$$

$$\begin{aligned} & \frac{31}{10} \\ & (1.55) \left(\frac{1}{10} \right) \left(\frac{31}{10} \right) \\ & \Rightarrow \frac{4.6}{10} = 4\% \end{aligned}$$

$$\frac{31}{10}$$

$$(SI - CI)_{3 \text{ yrs}} = \frac{P \times R^2 \times (300 + R)}{(100)^3}$$

$$15.5 = \frac{P \times \cancel{100} \times (310)}{\cancel{100} \times 100^2}$$

$$15.5 = \frac{310P}{100 \times 100}$$

$$1550 = \frac{310P}{100} \Rightarrow 1550 \times 100 = 310P$$

$$155 \times 10 = 310P \Rightarrow \frac{15500}{310} = P$$

$$\boxed{P = 500} \text{ Ans.}$$

17) The Compound Interest on Rs. 30,000 at 7% per annum is Rs. 4347. The period (in years) is:

Solⁿ $P = 30,000$, $R = 7\%$, $T = ?$, $CI = 4347$

first find SI (opt attach.)

- A) 2
B) 2.5
C) 3
D) 4
- w.r.t
SI $\Rightarrow$ Small
CI $\rightarrow$ Big

$$SI = \frac{PTR}{100} \text{ put } T = 2 \text{ yrs.}$$

$$SI = \frac{30000 \times 2 \times 7}{100} = 4200$$

$$2 \text{ yrs} = 4200 -$$

$$1 \text{ yrs} = 2100$$

$$0.5 \text{ yrs} = 1050 -$$

$$SI \text{ for } 2.5 \text{ yrs}$$

$$\begin{array}{r} 4200 \\ 1050 \\ \hline 5250 \end{array}$$

big than
CI

$$\therefore SI = 4200$$

is small

$$\boxed{\therefore CI = 2 \text{ yrs}}$$

Ans.

18) Find the C.I. on Rs. 5000 at 4% per annum for 1.5 years, compounded half yearly

① $5000 \xrightarrow{2\%} 100$

② $5100 \xrightarrow{2\%} 102$

③ $5202 \xrightarrow{2\%} 104.04$
306.04

$\frac{5202 \times 2}{100} = 104.04$

2×52.02

* Compounded

Half yearly	Quarterly
$\frac{R}{2}, t \times 2$	$\frac{R}{4}, t \times 4$
Ex: 1- $R \rightarrow 4\%, t \rightarrow 1.5$	Same
$\frac{4}{2}, 1.5 \times 2$	
$R \rightarrow 2\% \& 3 \rightarrow T$	

19) If the compound interest on a certain sum for 2 yrs at 10% p.a. is Rs 2100. What is the simple interest on it at the same rate for 2 yrs?

CI = 2100

R = 10%

T = 2 yrs

SI = ?

R = 10%

T = 2 yrs

SI = $\frac{PRT}{100}$ P = ?

SI = $\frac{10,000 \times 10 \times 2}{100}$

SI = 2000

Ans.

w.k.r

CI = $P \left(1 + \frac{R}{100} \right)^n - P$

$2100 = P \left[\left(1 + \frac{10}{100} \right)^2 - 1 \right]$

$2100 = P \left[\left(\frac{11}{10} \right)^2 - 1 \right]$

$2100 = P \left(\frac{121}{100} - 1 \right)$

$2100 = P \left(\frac{121}{100} - 1 \right)$

$P = \frac{2100 \times 100}{21}$

P = 10,000

20) A Sum is being lent at 20% per annum C.I. What is the ratio of the increased amount of the 4th year to the 5th year?

Ans

$$A = P \left(1 + \frac{r}{100} \right)^n$$

for ratio

$$\frac{\text{4th year}}{\text{5th year}} = \frac{P \left(1 + \frac{20}{100} \right)^4}{P \left(1 + \frac{20}{100} \right)^5} = \frac{1}{1 + \frac{20}{100}}$$

$$\frac{1}{\frac{120}{100}} \Rightarrow \frac{100}{120} = \frac{5}{6} = 83.33\%$$

Ans

Prp

21) A T.V. Set is available for Rs. 19,650 cash or Rs. 3100 down payment with three equal annual installments. If the shopkeeper charges 10% interest compounded annually, calculate the amount of each installment.

$$TV = 19650 \text{ [cash]} \quad A = P \left(1 + \frac{r}{100} \right)^n$$

for installment

$$dp = 3100$$

$$\text{Remaining } 16550 \rightarrow \text{paid in installment}$$

$$\frac{A}{1 + \left(\frac{r}{100} \right)^n} = P$$

$$\frac{1.21}{1.21} = 1$$

$$\frac{x}{1 + \frac{10}{100}} + \frac{x}{\left(1 + \frac{10}{100} \right)^2} + \frac{x}{\left(1 + \frac{10}{100} \right)^3} = 16550$$

$$\frac{x \times 10^{121}}{11} + \frac{100x^{11}}{121} + \frac{1000x^{11}}{1331} = 16550$$

$$\frac{1210x + 1100x + 1000x}{1331} = 16550$$

$$\frac{3310x}{1331} = 16550$$

$$\begin{array}{r} 1210 \\ 1100 \\ 1000 \\ \hline 3310 \end{array}$$

$$3310x = 16550 \times 1331$$

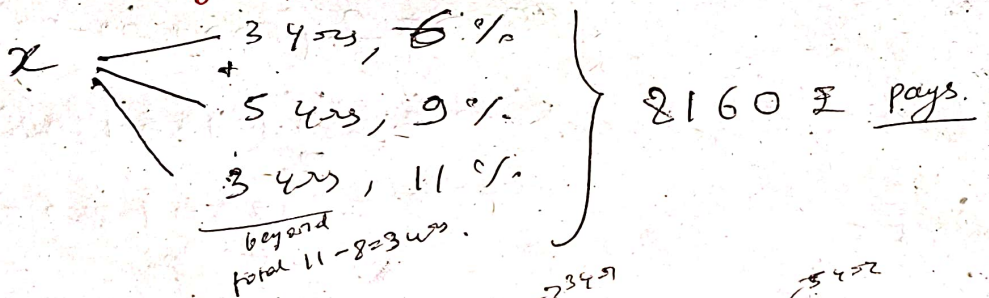
$$x = \frac{16550 \times 1331}{3310}$$

$$x = 6655$$

Ans

Imp

22) John borrowed some money at 6% p.a. for 3 yrs, 9% p.a. for the next 5 yrs, & 13% p.a. beyond 8 yrs. If John pays a total of Rs. 8160 after 11 yrs, how much did he borrow?



$$SI = \frac{PRT}{100}$$

$$\frac{x \times 6 \times 3}{100} + \frac{x \times 9 \times 5}{100} + \frac{x \times 13 \times 3}{100} = 8160$$

$$\frac{18x + 45x + 39x}{100} = 8160$$

$$\frac{102x}{100} = 8160$$

$$x = \frac{8160 \times 100}{102}$$

$$x = 8000$$

Ans

23) How long will it take to pay off a debt of Rs. 8800 if Rs. 250 is paid in the first month, Rs. 270 in the second month; Rs. 290 in the third month, and so on?

Q 23

1st month = 250

2nd month = 270

3rd month = 290

4th month =

5th month =

Arithmetic progression (AP).

Gap 220 $\Rightarrow$ d.

Let n =

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

A) 20 month

B) 24 m

C) 18 m

D) = 20.

$$S_n = 8800$$

$$8800 = \frac{n}{2} (2 \times 250 + (n-1)20)$$

$$10n^2 + 480n = 8800$$

put $n = 20$

$$10 \times 400 + 480 \times 20 = 8800$$

$$4000 + 9600 = 8800$$

$$\frac{n}{2} (480 + 20n) = 8800$$

put $n = 20$ m

$$\frac{20}{2} (480 + 20 \times 20) = 8800$$

$$4800 + 4000 = 8800$$

$$8800 = 8800$$

$$n = 20 \text{ month}$$

Ans,

$$8800 = \frac{n}{2} (500 + 20(n-1))$$

$$8800 = \frac{n}{2} (500 + 20n - 20)$$

$$8800 = \frac{n}{2} (480 + 20n)$$

$$8800 = 480n + 10n^2$$

$$8800 = 480n + 10n^2$$

10n²

24) In how many years will a Sum of money double itself at 12% per annum?

$$T = ?$$

$$P = x \rightarrow 2x - (\because P \text{ increase called Amount})$$

$$R = 12\%$$

W.K.T

$$A = P + I$$

$$2x = x + I$$

$$I = x$$

$$T = \frac{SI \times 100}{PR}$$

$$T = \frac{x \times 100}{x \times 12}$$

$$T = \frac{25}{3} = 8 \frac{1}{3}$$

$$T = 8 \text{ yrs } 4 \text{ months}$$

Ans

25) The ratio of the principle & its yearly amount is 8:9. What is the rate of simple interest per annum?

$$P = 8x$$

$$A = 9x$$

$$R = ?$$

$$R = \frac{SI \times 100}{PT}$$

$$R = \frac{x \times 100}{8x \times 1}$$

$$I = 9x - 8x$$

$$I = x$$

$$R = \frac{50}{4} = \frac{25}{2} = 12.5\%$$

Ans